

BIOMETRIC AUTHENTICATION FRAMEWORK FOR SECURE LIFETIME PATIENT HISTORY ACCESS

Project Report Summary | Sri Krishna College of Technology, Coimbatore

Ranjithraj T • Saran Selvakumar • Senthilnathan N

Department of Electronics and Communication Engineering | Anna University, Chennai | March 2026

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Executive Summary

Healthcare institutions increasingly rely on digital systems to manage patient records, but fragmented databases, weak authentication, and information overload continue to compromise both security and speed of care. This project presents a Biometric Authentication Framework for Secure Lifetime Patient History Access — a centralized system that layers RFID-based doctor verification, fingerprint and facial recognition for patients, and OTP-based emergency access on top of a NodeMCU-driven IoT hardware backbone. Record integrity is enforced through SHA-256 cryptographic hashing in a blockchain-inspired ledger, while an AI-powered summarization module condenses lengthy medical histories into concise clinical overviews. Prototype testing shows a 98.4% authentication accuracy, 1.2-second average response time, 100% data-integrity verification, and a 91% accurate AI summarization module, together representing substantial gains over conventional hospital record systems.

Project Overview

The framework combines IoT hardware, layered biometric identity verification, and intelligent software to deliver secure, efficient healthcare data management. A NodeMCU microcontroller coordinates an RFID reader, fingerprint sensor, and camera module, forwarding verified authentication data to a centralized server. Once authenticated, medical staff can retrieve or update patient records, which are protected by SHA-256 hashing for tamper detection and reviewed through an AI-generated summary before being presented to the user. The design targets both large hospital networks and smaller, resource-constrained clinics, and is intended to support future integration with wearable monitoring, national digital-health identity systems, and telemedicine platforms.

Problem Statement

Despite widespread digitization, hospital data systems continue to suffer from three linked problems. Patient records are dispersed across disconnected departmental databases, producing duplicate and inconsistent entries and slowing retrieval exactly when speed matters most. Authentication commonly still relies on passwords, ID cards, or PINs, which are vulnerable to phishing, credential theft, and insider misuse. Finally, the sheer volume of accumulated patient history — diagnoses, prescriptions, lab results, and notes — makes manual review slow and increases the risk that clinicians overlook critical details during time-pressured decisions. Industry data cited in the report indicates that roughly 64% of hospitals face cross-department interoperability issues, 43% of healthcare data breaches trace back to weak authentication, and about 20% of records in non-integrated systems contain duplicate or mismatched entries.

Objectives

- Build a secure, centralized system for lifetime storage and retrieval of patient records across multiple healthcare sources.
- Implement biometric authentication (fingerprint and facial recognition) for tamper-proof patient identification.
- Verify doctors through RFID-based access control so only authorized personnel can reach patient data.
- Provide an OTP-based emergency access path requiring both patient and guardian authorization.
- Preserve data integrity and privacy using blockchain-inspired SHA-256 encryption.

- Integrate an AI/LLM-based summarization module for faster, more informed clinical decisions.
- Ensure scalability, reliability, and interoperability, with alignment to HIPAA/GDPR-style data protection standards.

Existing System

Conventional electronic health record (EHR) setups store data in siloed, department-specific systems with limited interoperability, forcing repeated data entry and slowing access during procedures. Authentication is typically password- or card-based, which is easy to compromise, and there is little audit-level traceability of who accessed or changed a record. Legacy infrastructure is also rigid, making it difficult to add capabilities such as remote access or AI-assisted analysis, and existing systems generally lack any automated way to condense long patient histories for clinicians.

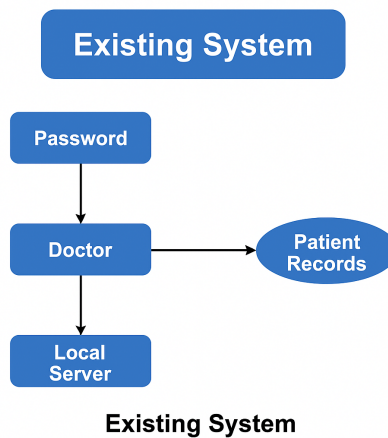


Fig. 1 — Existing EHR workflow: departmental segregation and limited authentication security.

Proposed System

The proposed framework replaces fragmented, password-based access with a layered, hardware-backed verification pipeline feeding a single centralized database. Doctors authenticate via RFID; patients confirm identity through fingerprint and facial recognition; emergencies fall back to dual OTP authorization for patient and guardian. A NodeMCU microcontroller manages the sensors and secures communication with the backend server, where SHA-256 hashing protects every record and an AI/LLM summarization engine produces a concise clinical brief on request. The result is a system that is simultaneously harder to breach and faster to use than the systems it replaces.

System Architecture

The architecture is organized in two cooperating layers. The hardware layer — NodeMCU, RFID reader, fingerprint sensor, camera module, and GSM/Wi-Fi communication — captures authentication data and relays it securely to the server. The software layer — backend server, database, and AI summarization module — authenticates requests, stores and retrieves records, and generates summaries. Authentication is multi-stage: doctor RFID verification precedes patient biometric verification, after which records are pulled from the centralized database, passed through the AI summarizer, and returned to the requesting user. All hardware-to-server traffic is encrypted, every record is SHA-256 hashed, and all activity is logged for accountability, giving the design a straightforward path to scale as additional sensors or facilities are added.

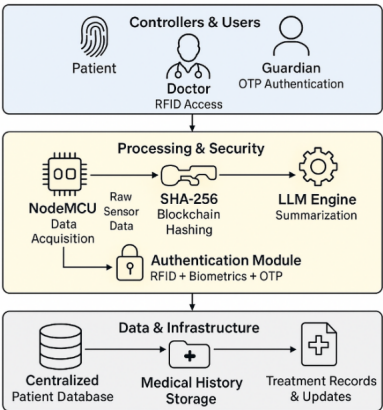


Fig. 2 — System architecture / representation diagram (original, Fig. 3.1 of the source report).

Hardware Components

Component	Role
NodeMCU Microcontroller	Central processing unit; interfaces with sensors and manages server communication.
RFID Reader & Tags	Authenticates doctors by reading encrypted credentials from authorized cards.
Fingerprint Sensor (R305 / GT521F52)	Captures and matches patient fingerprints for biometric authentication.
Camera Module (OV7670)	Performs facial recognition for multimodal patient verification.
GSM / Wi-Fi Module	Provides network connectivity and delivers OTPs for emergency access.

Software & Development Tools

- **Python 3.10.10** — core language for backend and AI logic; used with cryptography, API, and ML libraries such as TensorFlow / Hugging Face for the summarization feature.
- **Thonny IDE** — primary Python development/debugging environment, chosen for its simplicity and integrated debugger.
- **Flask** — lightweight web framework exposing RESTful APIs for login verification, biometric validation, OTP handling, and record transmission.
- **Arduino IDE** — used to program and test the NodeMCU firmware for the RFID and fingerprint modules.
- **React.js** — powers the web-based dashboard for record management, data entry, and viewing AI-generated summaries.

Module-wise Explanation

Doctor Authentication (RFID)

Medical staff are issued encrypted RFID tags. The NodeMCU reads the tag over SPI and cross-checks the credential against the central database before granting session access — replacing shared passwords with a physical, hard-to-clone credential.

Patient Verification (Fingerprint & Facial Recognition)

A fingerprint sensor communicates with the NodeMCU over UART, extracting ridge and bifurcation features into a digital template that is matched against stored templates. A camera module adds facial recognition as a second biometric factor, together forming multimodal verification that is difficult to spoof.

Emergency Access (OTP)

When biometric verification is unavailable in an emergency, the system issues a one-time password over GSM/Wi-Fi to the patient's registered number, requiring both patient and guardian confirmation before access is granted — trading some convenience for continued auditability.

Data Integrity (SHA-256)

Every medical record is hashed with SHA-256 before storage. Any later modification changes the computed hash, immediately revealing tampering and preserving a blockchain-inspired, tamper-evident audit trail.

Patient Enrollment

New patients are registered with core demographic data (name, age, gender, phone number, blood group), after which staff trigger a face-capture step to bind biometric identity to the record, preventing duplicate or fraudulent registrations.

Authentication Workflow (RFID, Fingerprint, OTP, SHA-256)

- **1. Doctor login:** RFID card is scanned; NodeMCU validates the tag against the database.
- **2. Patient verification:** fingerprint scan (and/or facial recognition) confirms patient identity.
- **3. Emergency fallback:** if biometric verification is unavailable, an OTP is sent for dual patient + guardian authorization.
- **4. Integrity check:** the requested record's stored SHA-256 hash is verified before release, confirming no tampering has occurred.
- **5. Retrieval & summarization:** the verified record is pulled from the centralized database and passed to the AI summarizer.
- **6. Logged delivery:** the summarized record is shown to the user, with the entire transaction timestamped for audit.

Database Design

The database is organized around four core tables. The **Patient** table holds demographic data (name, age, gender, phone, blood group) under a unique patient ID. The **Doctor** table stores doctor ID, RFID card number, and department for use during RFID authentication. The **Medical Records** table holds diagnoses,

prescriptions, and lab results, each entry timestamped and paired with a cryptographic hash for integrity verification. The **Authentication Log** table records every login attempt — successful or failed — along with biometric and OTP verification outcomes, supporting security monitoring. Indexing and encryption are applied throughout to keep retrieval fast while protecting stored data from unauthorized access.

AI Summarization Module

Long-form patient histories — clinical notes, diagnostic reports, prescriptions — are processed with natural language processing to extract key medical concepts and relationships. The module retrieves relevant record sections from the database and generates a structured summary highlighting chronic conditions, current medications, allergies, and recent procedures, which is then shown to clinicians through the dashboard. In testing this reduced the time doctors spent reviewing patient histories by roughly 40%, while achieving 91% summarization accuracy, a 92% F1-score, and 93% precision.

Testing Methodology

Testing focused on validating each authentication factor individually and in combination, using the assembled NodeMCU hardware prototype programmed through the Arduino IDE. RFID testing checked that pre-registered cards were consistently accepted and unregistered cards rejected. Fingerprint testing covered enrolled users under varied conditions, including partial impressions and finger-placement variation, alongside verification that newly enrolled fingerprints were stored and retrievable without duplication. OTP testing confirmed correct delivery and validation, including rejection of invalid codes. System-level metrics — authentication accuracy, response time, data-integrity index, summarization accuracy, and cost efficiency — were then benchmarked against a conventional-system baseline (see Table below and Test Results).

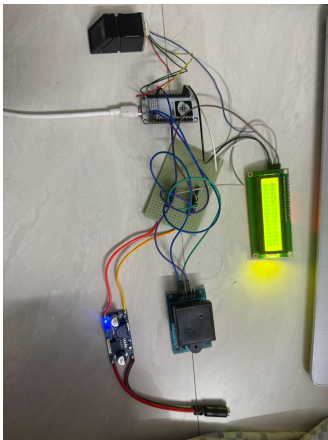


Fig. 3 — NodeMCU hardware prototype used for testing.

Test Results

Metric	Existing System	Proposed System	Improvement
Authentication Accuracy (%)	92	98.4	+6.4%
Response Time (s)	2.1	1.2	45% faster
Data Integrity Index (%)	87	100	+13%
Summarization Accuracy (%)	82	91	+9%
System Cost Efficiency (%)	65	88	+23%

The RFID and fingerprint modules were validated on the physical prototype (enrollment and login screenshots below); false acceptance stayed at 0.8% and false rejection below 2% across repeated trials.

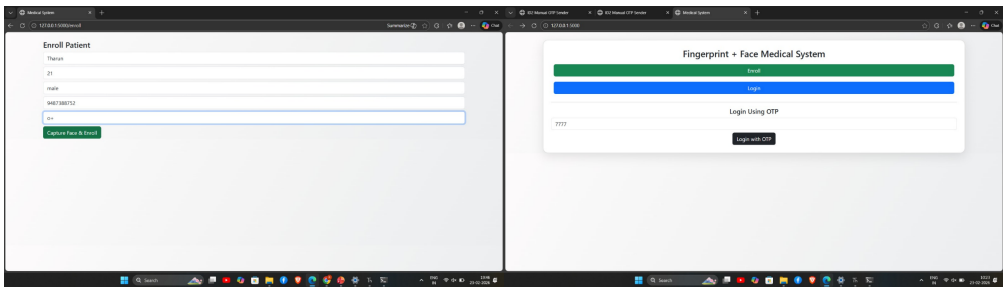


Fig. 4 — Patient enrollment interface.

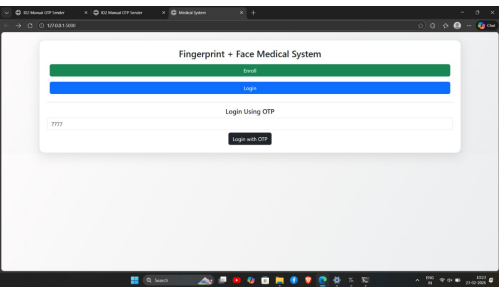


Fig. 5 — OTP login interface.

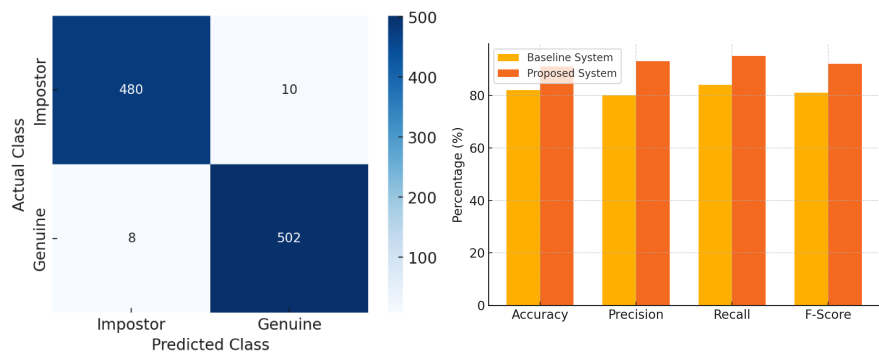


Fig. 6 — Confusion matrix: biometric classification accuracy.

Fig. 7 — AI summarization performance vs. baseline.



Fig. 8 — Overall performance comparison across evaluation metrics.

Advantages

- Multi-factor security (RFID + biometrics + OTP) sharply reduces impersonation and unauthorized access risk.
- Centralized architecture lets clinicians retrieve full patient histories instantly instead of searching multiple systems.
- AI summarization cuts record-review time by ~40%, helping in time-critical situations.
- SHA-256 hashing guarantees tamper-evident, verifiable medical records.
- Low-cost IoT hardware (NodeMCU) and open-source tools keep implementation and maintenance costs down (~23% cheaper).
- Modular design allows straightforward expansion to wearables, predictive analytics, and nationwide health-ID systems.

Applications

- Hospital and clinic record management, including emergency and outpatient departments.
- Telemedicine platforms, using biometric verification to secure remote consultations.
- Health insurance claim verification through authenticated access to treatment records.
- National digital-health identity initiatives requiring lifelong, cross-institution medical records.
- Rural and resource-limited healthcare settings, enabled by the low-cost, lightweight hardware footprint.
- Anonymized clinical research and public-health data analysis.

Future Enhancements

- Add advanced biometrics such as iris scanning or voice recognition for further accuracy gains.
- Complete integration of facial recognition and OTP-based guardian verification (currently partially implemented).
- Move to a cloud-based centralized database for multi-facility, lifetime record access.
- Integrate wearable devices and IoT sensors for continuous health monitoring.
- Deepen blockchain-based storage for stronger immutability guarantees.

Conclusion

The Biometric Authentication Framework demonstrates a workable path toward secure, scalable, and intelligent healthcare data management. By combining multi-factor authentication, blockchain-inspired data validation, and AI-driven summarization, it addresses the fragmentation, weak security, and information-overload problems of conventional systems. Prototype results — 98.4% authentication accuracy, 1.2-second average response time, 100% data-integrity verification, and a 91%-accurate summarization module — indicate meaningful gains over existing baselines, while the low-cost hardware and modular design keep the approach viable for both small clinics and larger hospital networks. The current prototype covers roughly 70% of the planned scope; remaining work centers on facial recognition, OTP-based guardian verification, and full AI summarization integration, after which the framework would be positioned as a complete, deployable healthcare data platform.